

# Biological Foundations of Development



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## Nature or Nurture?

### Nature

- Genetic factors/heredity
- Stability
  - Tendency towards sameness over time.

### Nurture

- Environmental factors
- Plasticity
  - Likelihood of change if environment is supportive

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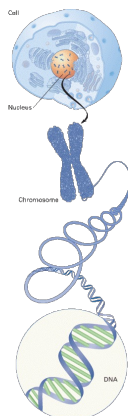
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## Science of Genetics

The study of heredity, the inborn factors inherited from biological parents which affect development.



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## GENETIC FOUNDATIONS

### Genotype

- The genetic makeup of an individual
- supplied by heredity

### Phenotype

- An individual's observable physical and behavioural characteristics
- can be determined by both genetic and environmental factors

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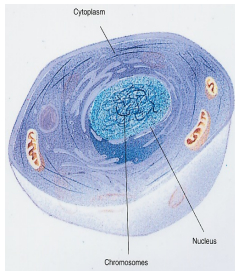
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## A Human Cell



- Each cell of human body contains a nucleus
- Nucleus contains chromosomes.

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## Chromosomes

- Rod-like structures in the cell's nucleus
- Store and transmit genetic information
- Can be arranged in pairs.



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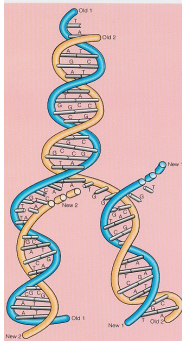
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## Deoxyribonucleic Acid (DNA)



- Chromosomes are made up of **DNA**.
- DNA can duplicate itself by **mitosis**.

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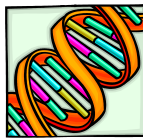
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## Genes

- Genes are segments of DNA located along the chromosomes.
- Genes contain hereditary material passed from parents to children.



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**22 pairs** of matching chromosomes are called **autosomes**.

**23rd pair** is the **sex chromosomes**.

- Females have an **XX** pair.
- Males have an **XY** pair.



Normal Karyotype

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## The Sex Cells

The sperm in males and the ova in females.

- Contain 23 single chromosomes.
- Formed by a special type of cell division known as meiosis which halves the number of chromosomes.
- No two gametes are exactly alike - ensures variability in the species.

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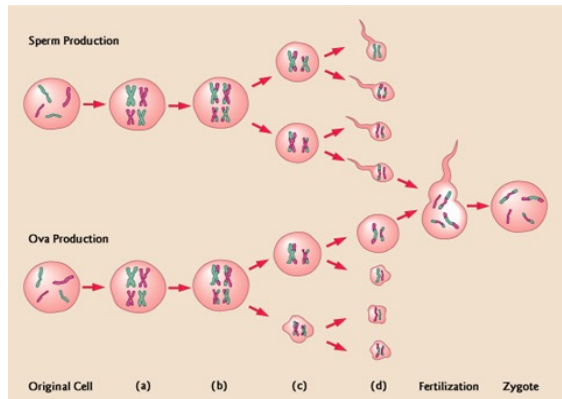
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## Meiosis



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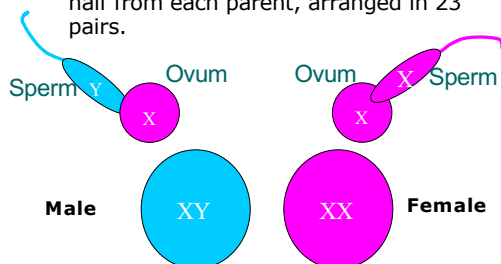
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## Sex: Male or Female?

- An X- or Y-bearing sperm fertilises the ovum.
- Resulting zygote has 46 chromosomes, half from each parent, arranged in 23 pairs.



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## Multiple Births

- **Identical or monozygotic**

- A single zygote separates to form two or more identical individuals.

- **Fraternal or dizygotic**

- Two separate ova are fertilised by two different sperm.



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## Patterns of Genetic Inheritance

- If the genes from mother and father are alike, the inherited trait will be displayed
- If they are different, then the relationships between the genes determine the trait that will appear.

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## Genetic principles: Single gene pairs

- **Dominant:** a single affected gene causes the trait to be expressed (e.g. Huntington's Disease)
- **Recessive:** two affected genes needed for the trait to be expressed (e.g., cystic fibrosis). Carriers have one affected gene overridden by normal gene.
- **Sex-linked:** Affected gene carried on X chromosome (e.g., colour blindness, haemophilia).

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- **Polygenic inheritance**

Many interactive pairs of genes are involved in determining complex traits (e.g., height, IQ).

- **Chromosomal Abnormalities**

Whole or large sections of chromosome affected (e.g., Down's Syndrome extra 21<sup>st</sup> chromosome).

- **Abnormalities of sex chromosomes**

Missing/extra/damaged sex chromosomes (e.g., XO Turner's syndrome, Fragile X syndrome).

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### **The relationship between heredity and environment**

- **Behaviour genetics**

- Twin studies, adoption studies look at the relative influence of genes and environment
- The most famous is the Minnesota Twin Study begun in 1979 by Tom Bouchard
  - See the link in Topic 4 in Stream

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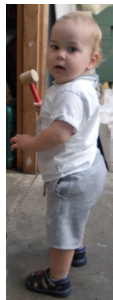
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### **Genetic-Environment Correlation**

The extent to which heredity influences the environments to which we are exposed.



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## Genetic-Environment Correlation

- **Passive**  
Caregivers control child's environment
- **Evocative**  
Child's behaviour is consistent with own heredity, reinforced by others.
- **Active**  
Older children actively select environments that fit with their genetic tendencies (*niche-picking*).

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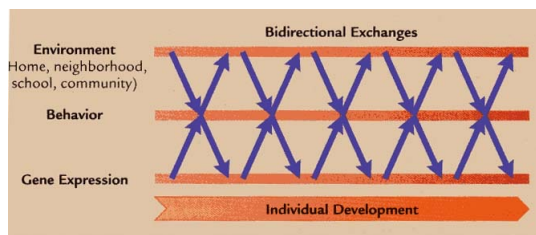
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## The Epigenetic Framework



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## Prenatal Development and Birth

- Germinal period
- Embryonic period
  - Critical period
  - Teratogens
- Foetal period



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## Periods of Prenatal Development

| Period           | Weeks            | Key Events                                                                                                                        |
|------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Germinal</b>  | first 2 weeks    | <ul style="list-style-type: none"> <li>■ Fertilisation</li> <li>■ Implantation</li> <li>■ Start of Placenta</li> </ul>            |
| <b>Embryonic</b> | 2-8 weeks        | <ul style="list-style-type: none"> <li>■ Arms, legs, face, organs, muscles all develop</li> <li>■ Heart begins beating</li> </ul> |
| <b>Foetal</b>    | 9 weeks to birth | <ul style="list-style-type: none"> <li>■ "Growth and finishing"</li> </ul>                                                        |

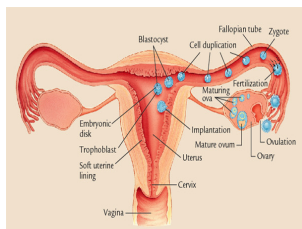
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- Physical growth and development follows two organised patterns:

- **Cephalocaudal** (head to tail)
- **Proximodistal** (near to far, centre outwards)

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## Germinal period Fertilisation to 2 weeks



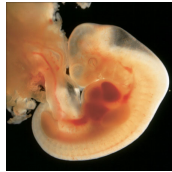
- Conception occurs in fallopian tube to form one-celled **zygote**.
- By 4<sup>th</sup> day forms a **blastocyst** - fluid-filled ball
- Implantation:
  - Inner cells will become the foetus.
  - Outer cells called the **trophoblast** will form yolk sac, membranes, placenta, and umbilical cord.

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### Embryonic period

3rd to 8th week

- **Period of Organogenesis:**  
all the major organs are formed
- Embryo forms three layers in week 3:
  - **Endoderm**
    - Digestive and respiratory system
  - **Ectoderm**
    - Nervous system and skin
    - The **neural tube**, **neurogenesis**
  - **Mesoderm**
    - Muscles, skeleton, circulatory system



5 weeks

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### Embryonic period

– 2<sup>nd</sup> month

- Eyes, ears, nose, jaw and neck form
- Buds become arms, legs, fingers and toes
- Internal organs continue to develop.



6 weeks



8 weeks

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### Foetal period

9 weeks till birth

- **End of First Trimester**
  - 9<sup>th</sup> to 12<sup>th</sup> week: External genitals are formed, sex evident
  - Organs and systems becoming organised and connected
  - Fingernails, toenails, tooth buds, eyelids appear
- **Second Trimester**
  - 17<sup>th</sup> to 20<sup>th</sup> week: Quickening
  - Most of brain's neurons in place
    - **neuronal migration**



Week 15



Week 19

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Foetal period  
9 weeks till birth



- **Third Trimester**

- Age of viability 22-26 weeks
- Growth slows and body systems completed
  - Cerebral cortex enlarges, neural connectivity begins.
  - Fat layer develops
- Can hear, feel, learn and remember

Week 26

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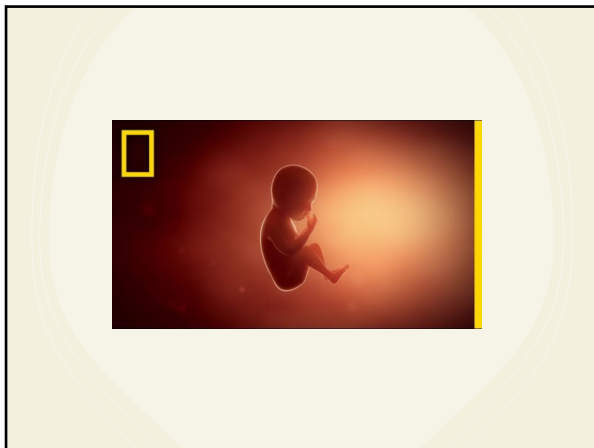
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### Critical Period

- “A time-limited period during which certain developmental changes are highly vulnerable to disruption” (Hoffnung et al., 2016, p.87)
- Teratogens: “Any agent that can potentially cause a birth defect or negatively alter cognitive and behavioural outcomes” (Santrock, 2018, p.56)

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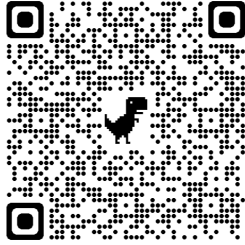
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### Activity



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### Other Maternal Factors which may influence prenatal development

- Age
- Nutrition
- Emotional Stress



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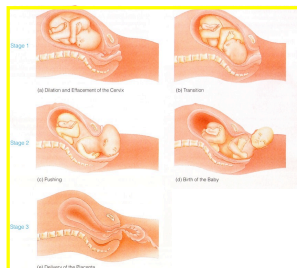
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### STAGES OF CHILDBIRTH

- **Dilation and effacement of the cervix**
  - On average, 6 to 12 hours with a first baby
- **Delivery of the baby**
  - 50 minutes for a first baby and 20 minutes in subsequent births
- **Afterbirth**
  - Expulsion of the placenta



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**Definitions:** see if you /group can define these terms

epigenetic view  
genes  
genotype  
phenotype  
meiosis  
mitosis  
behavioural genetics  
teratogens

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